Searching Algorithms Programs in C

1.Program to Implement Binary Search Using C

```
#include<stdio.h>
#include<conio.h>
void main(){
   int a[10], n, i, low, high, mid, key, found=0;
   clrscr();
   printf("Enter number of elements: ");
   scanf("%d",&n);
   printf("Enter sorted elements:\n");
   for(i=0;i<n;i++){
   scanf("%d",&a[i]);
   }
   printf("Enter element to search: ");
   scanf("%d",&key);
   low=0;
   high=n-1;
   while(low<=high){
      mid=(low+high)/2;
      if(a[mid]==key){
         found=1;
         break;
      }
      else if(a[mid]<key){
         low=mid+1;
      }
      else{
         high=mid-1;
```

```
        }
    }
    if(found)
        printf("Element found at position %d",mid+1);
    else
        printf("Element not found");
    getch();
}
```

Output:

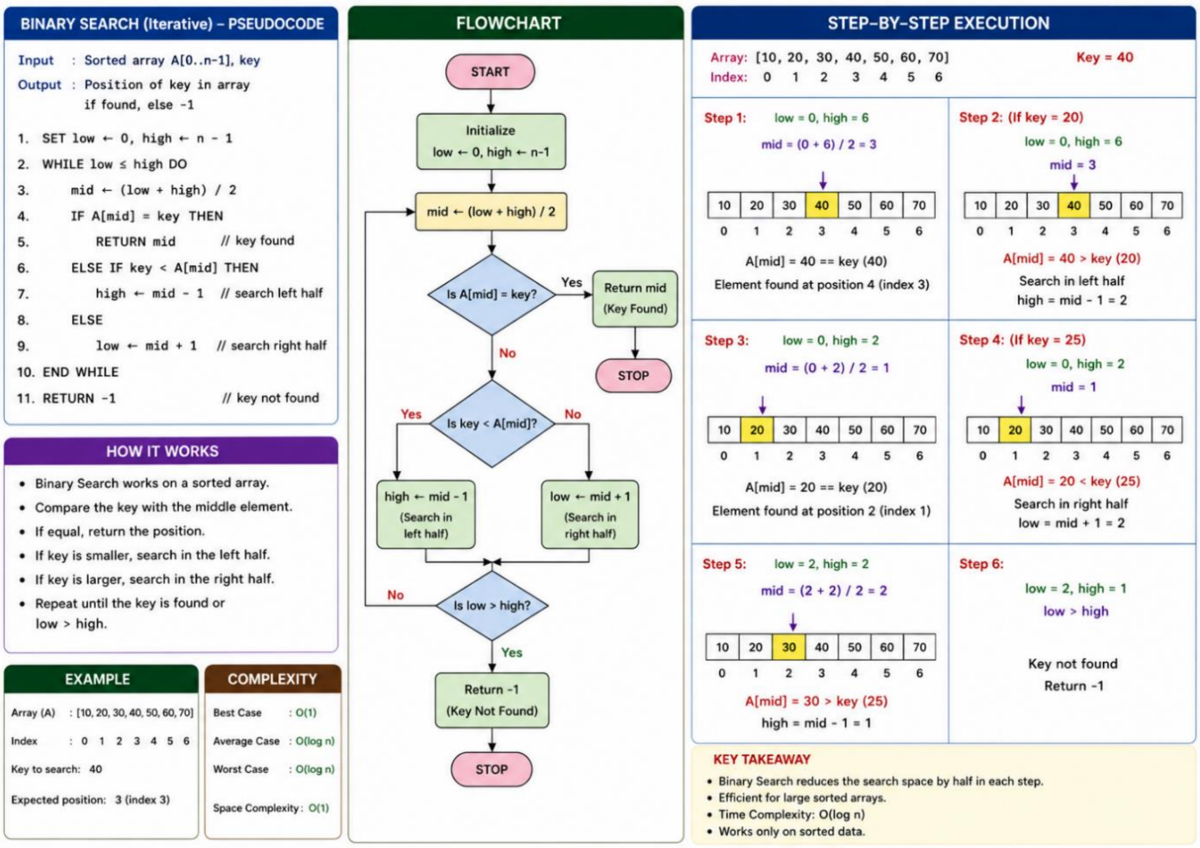
Enter number of elements: 5

Enter sorted elements:

10 20 30 40 50

Enter element to search: 30

Element found at position 3



2. Program to Implement Linear Search Using C

```
#include<stdio.h>
#include<conio.h>
void main(){
   int a[100], n, i, key, found = 0;
   clrscr();
   printf("Enter number of elements: ");
   scanf("%d", &n);
   printf("Enter elements:\n");
   for(i=0; i<n; i++){
      scanf("%d", &a[i]);
   }
   printf("Enter element to search: ");
   scanf("%d", &key);
   for(i=0; i<n; i++){
      if(a[i] == key){
         found = 1;
         break;
      }
   }
   if(found)
      printf("Element found at position %d", i+1);
   else
      printf("Element not found");
   getch();
}
```

Output: Enter number of elements: 5

Enter elements:

10 25 30 45 50

Enter element to search: 45

Element found at position 4